

Basic Principles of Medicine 2

Nervous System and Behavioral Science

Neurotransmitter Pathways and Roles

DLA (Part 1)

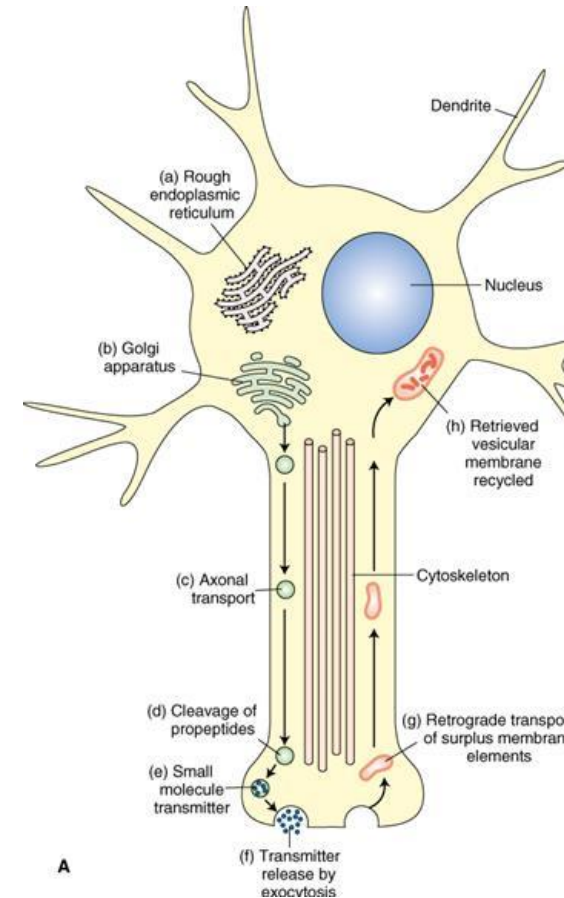
Learning Objectives

- SOM.MK.I.BPM2.1.NB.1.NSCI.0101 List the five traditional criteria that define a neurotransmitter
- SOM.MK.I.BPM2.1.NB.1.NSCI.0102 Classify neurotransmitters according to molecular weight
- SOM.MK.I.BPM2.1.NB.1.NSCI.0103 Describe mechanisms for synthesis of neurotransmitters
- SOM.MK.I.BPM2.1.NB.1.NSCI.0104 Describe mechanisms for loading of transmitters into vesicles
- SOM.MK.I.BPM2.1.NB.1.NSCI.0105 Describe the synthesis/removal of acetylcholine
- SOM.MK.I.BPM2.1.NB.1.NSCI.0106 Describe the synthesis/removal of amino acid transmitters
- SOM.MK.I.BPM2.1.NB.1.NSCI.0107 Describe the synthesis/removal of catecholamines and serotonin

How would you classify Neurotransmitters

Traditional Criteria for Neurotransmitters

- **Synthesized** by presynaptic neurons
- **Stored** in preparation for release
- **Released** by presynaptic neurons in a **calcium dependent** fashion
- Specialized **receptors** (typically postsynaptic) effect stereotyped transmitter and analogue dependent physiological changes
- Mechanism for **removal** of transmitter from synapse

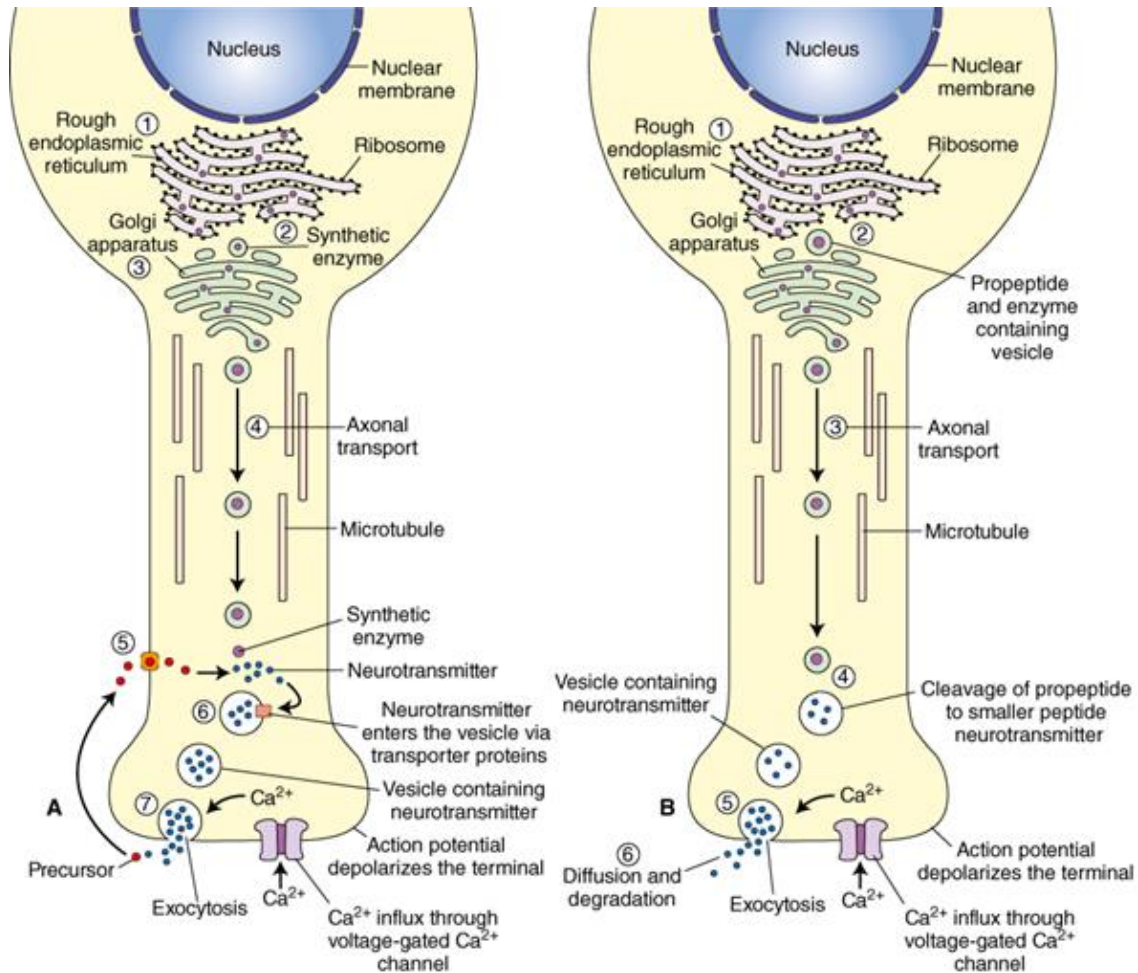


Major Classes of Neurotransmitters

Small-Molecule Neurotransmitters	Neuropeptides	Gaseous Neurotransmitters
Acetylcholine Excitatory amino acids Glutamate Aspartate Inhibitory amino acids GABA Glycine	Opioid peptides β -Endorphin Enkephalins Nociceptin	Nitric oxide
Biogenic amines Catecholamines Dopamine Norepinephrine Epinephrine	Tachykinins: Substance P	
Indoleamine Serotonin (5-hydroxytryptamine [5-HT])		
Imidazole amine Histamine		
Purines ATP Adenosine		

High molecular weight neurotransmitters (neuroactive peptides) are vastly more numerous

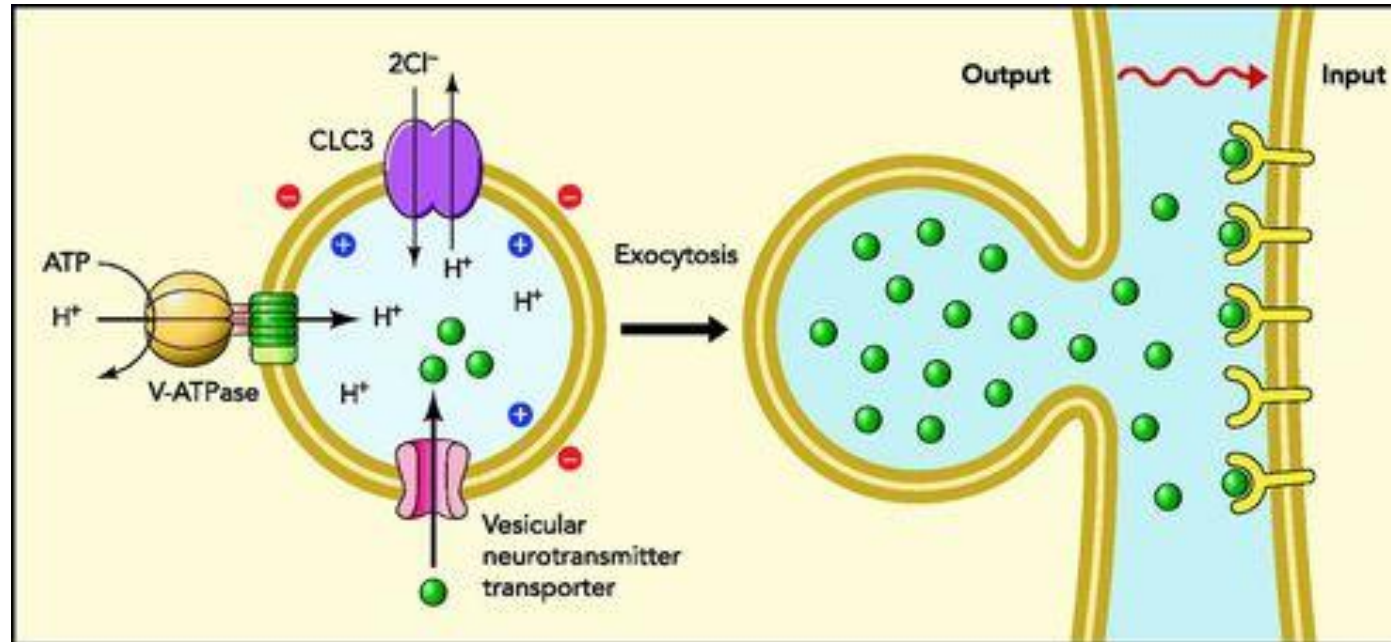
Steps involved in the synthesis, transport, and release of neurotransmitters



A. Small-molecule neurotransmitters. B. Neuropeptides

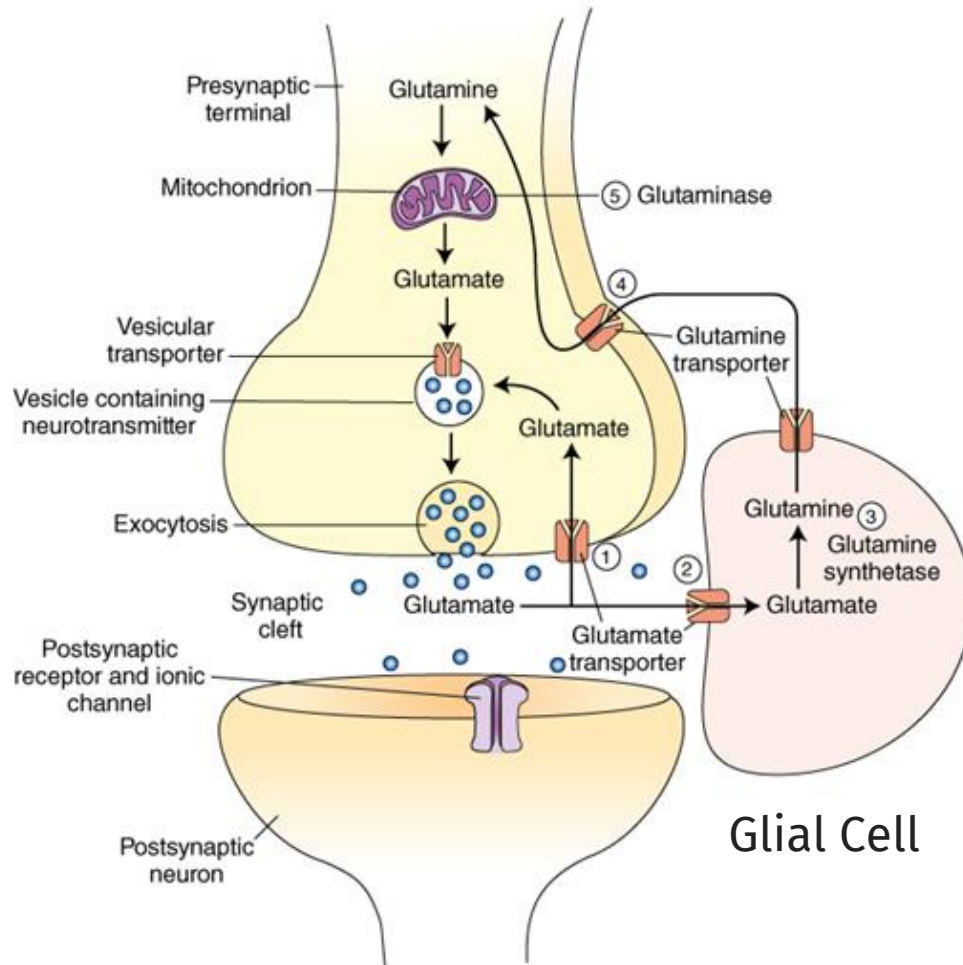
Properties	Small molecule neurotransmitter	Large molecule neurotransmitter
Size	Small (ex. Amino acid or amine)	Large (Ex. Polypeptide)
Synthesis	By uptake or by enzymes	By protein synthesis
Vesicle	Small (filled by the neurotransmitter)	Large (secreted proteins from rough endoplasmic reticulum)
Duration of action	Fast but short	Slow and long

Vesicular Loading



- Energy driven pump loads vesicles with hydrogen ions
- Antiport substitutes transmitter for hydrogen

Steps involved in the synthesis and release of glutamate



Steps involved in the synthesis and release of glutamate

Synthetic Pathway 1

Krebs (TCA) cycle

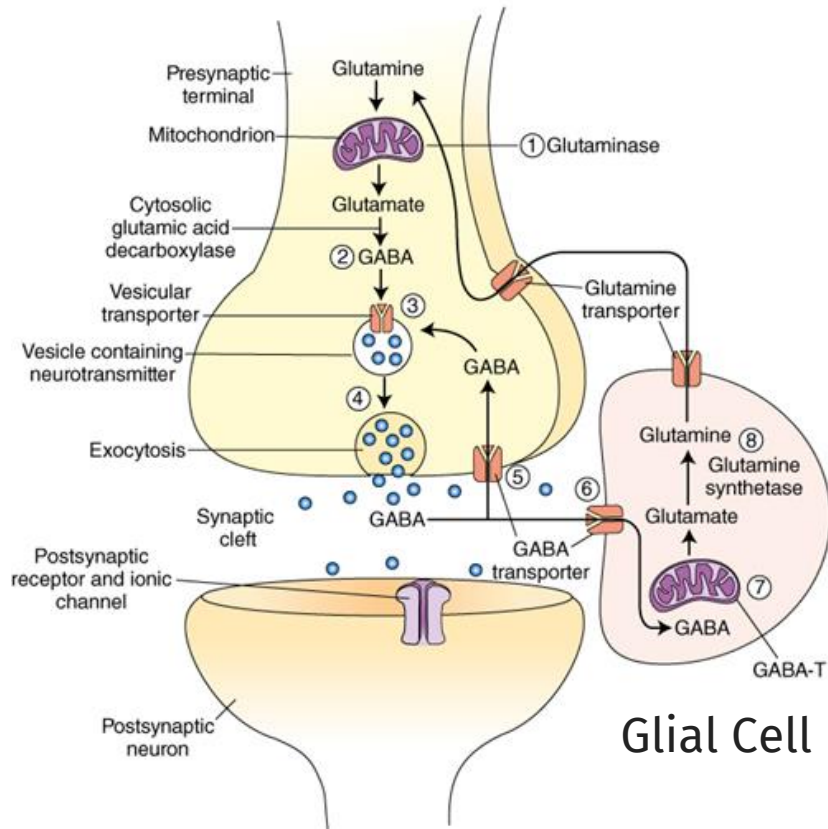
1. **Glucose** enters neuron by facilitated diffusion
2. Intracellular glucose is metabolized via Krebs cycle
3. **Alpha-oxoglutarate transaminase** yields glutamate

Synthetic Pathway 2

Glutamate recycling

1. Terminal and astrocytic glutamate transporters take up extracellular glutamate
2. Glutamine synthetase metabolizes glutamate to form glutamine in astrocytes
3. Glutamine exits astrocytes and enters neurons through glutamine transporters
4. Intraneuronal glutaminase converts glutamine to glutamate for reloading into vesicles
5. Glutamate taken up by neuronal terminals is also subject to vesicular reloading

Synthesis and Removal of Amino Acid Transmitters: GABA and Glycine



Steps involved in the synthesis and release of γ -aminobutyric acid (GABA)

Synthesis and removal of GABA

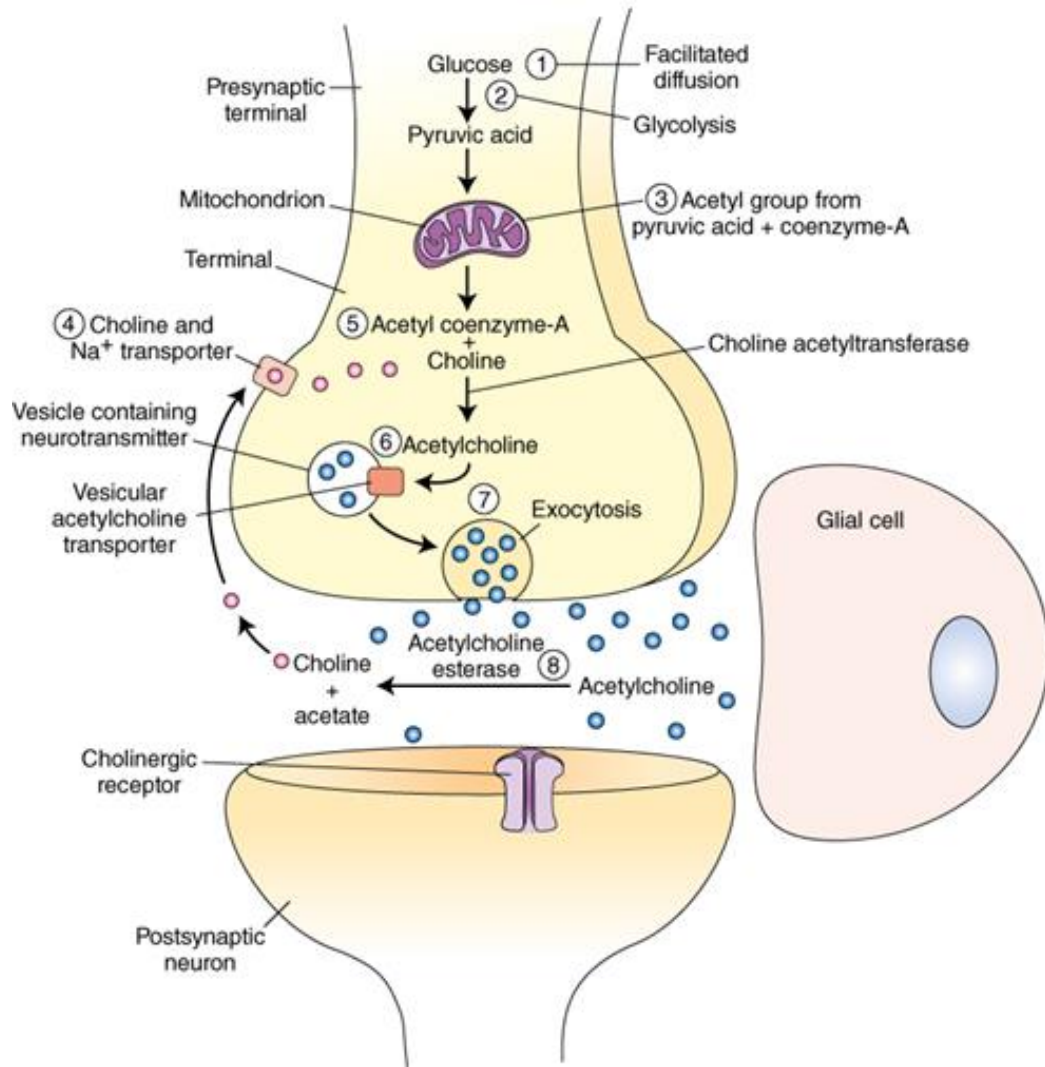
1. Glutaminase converts glutamine to glutamate
2. **Glutamic acid decarboxylase** converts glutamate to GABA*
3. GABA enters vesicles
4. After release, GABA transporters take up GABA for reuse
5. Glia take up GABA, where **GABA transaminase** degrades GABA to form glutamate
6. Glutamine synthetase then converts glutamate to glutamine
7. Glutamine return to neurons for re-synthesis of glutamate then GABA

Synthesis and removal of Glycine

1. Glycolysis of glucose yields **3phosphoglycerate** and subsequently serine
2. **Serine transhydroxymethylase** folate dependently converts serine to glycine
3. Membrane spanning transporters take up synaptic glycine

***Pyridoxylphosphate (active vitamin B6) acts as a coenzyme**

Synthesis and Removal of Acetylcholine



1. Glucose enters cell through facilitated diffusion

2. Cytoplasmic glycolysis generates pyruvate

3. Pyruvate enters mitochondria

- Donates acetyl group to coenzyme-A
- Acetyl coenzyme-A returns to cytoplasm

4.*Choline retrieved from the synapse interacts with acetyl coenzyme-A in presence of Ach transferase to yield ACh

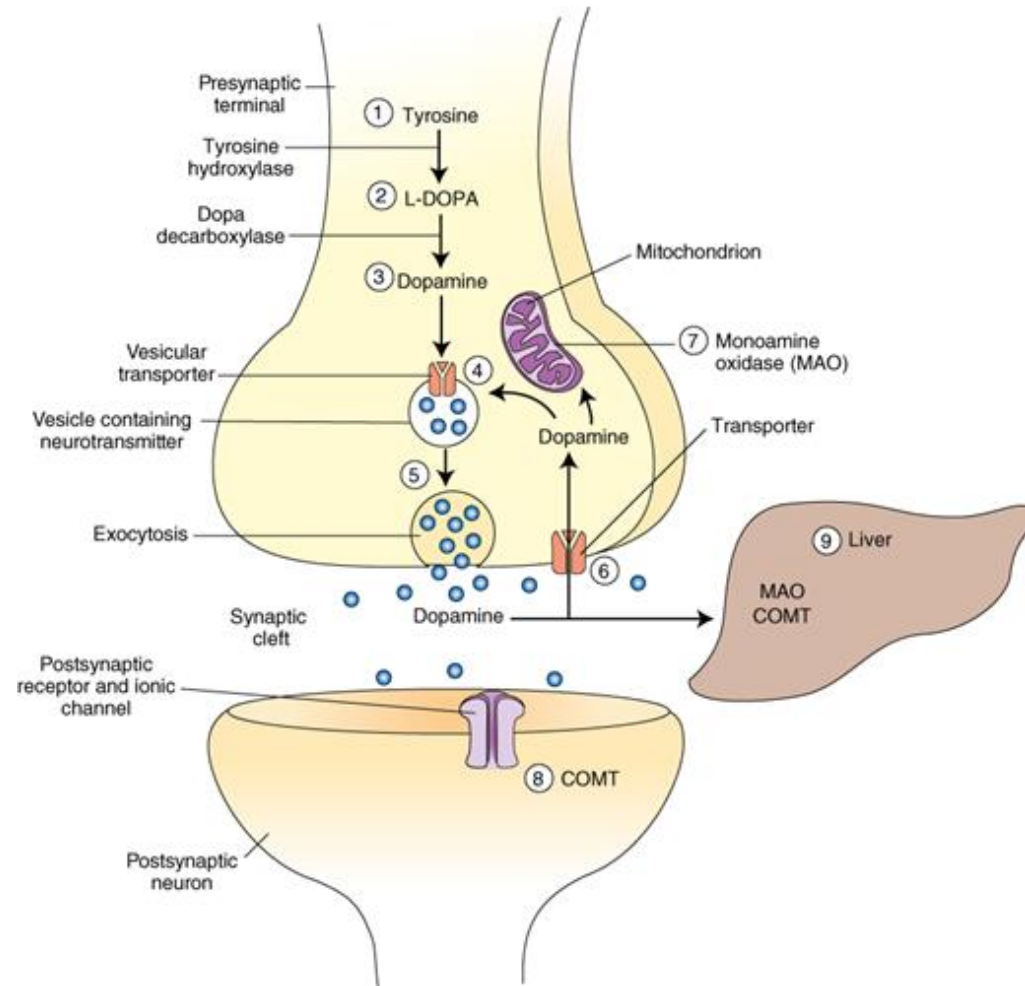
5. Ach enters vesicles

6. Ach-esterase hydrolyzes Ach (resultant choline taken up for re-use)

***Choline is present in plasma. Acetylcholine (Ach) formation is limited by the intracellular concentration of choline, which is determined by uptake of choline into the nerve ending.**

Steps involved in the synthesis and release of acetylcholine

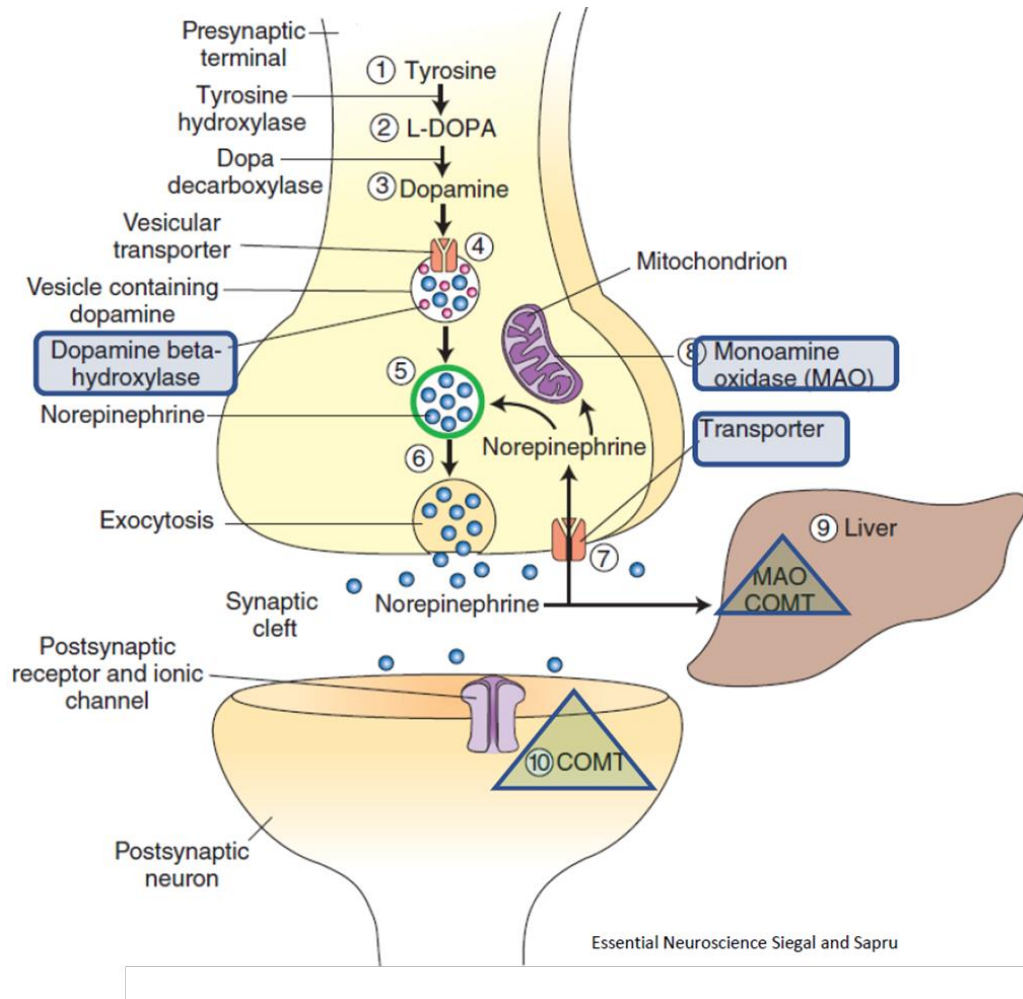
Synthesis and Removal of Catecholamine Transmitters: Dopamine



Steps involved in the synthesis and release of dopamine. *COMT*, catechol-O-methyltransferase *L-DOPA*, levodopa.

- Tyrosine actively transported into catecholaminergic neurons
- **Tyrosine hydroxylase** converts tyrosine to L-DOPA
- **Dopa decarboxylase** converts L-DOPA to Dopamine (DA)
- DA loaded into vesicles for release
- Reuptake-1: actively transports DA into presynaptic neuron
 - a. Some DA reloaded into vesicles
 - b. Remainder metabolized by MAO
- Reuptake-2: actively transports DA into postsynaptic cell for metabolism by COMT
- Remaining synaptic DA diffuses and is absorbed by blood for peripheral

Synthesis and Removal of catecholamines: Norepinephrine



1. DA enters vesicles (as above) with **dopamine- β -hydroxylase**, which converts DA to NE

2. Reuptake-1: actively transports NE into presynaptic neuron

- Some NE reloaded into vesicles
- Remainder metabolized by MAO

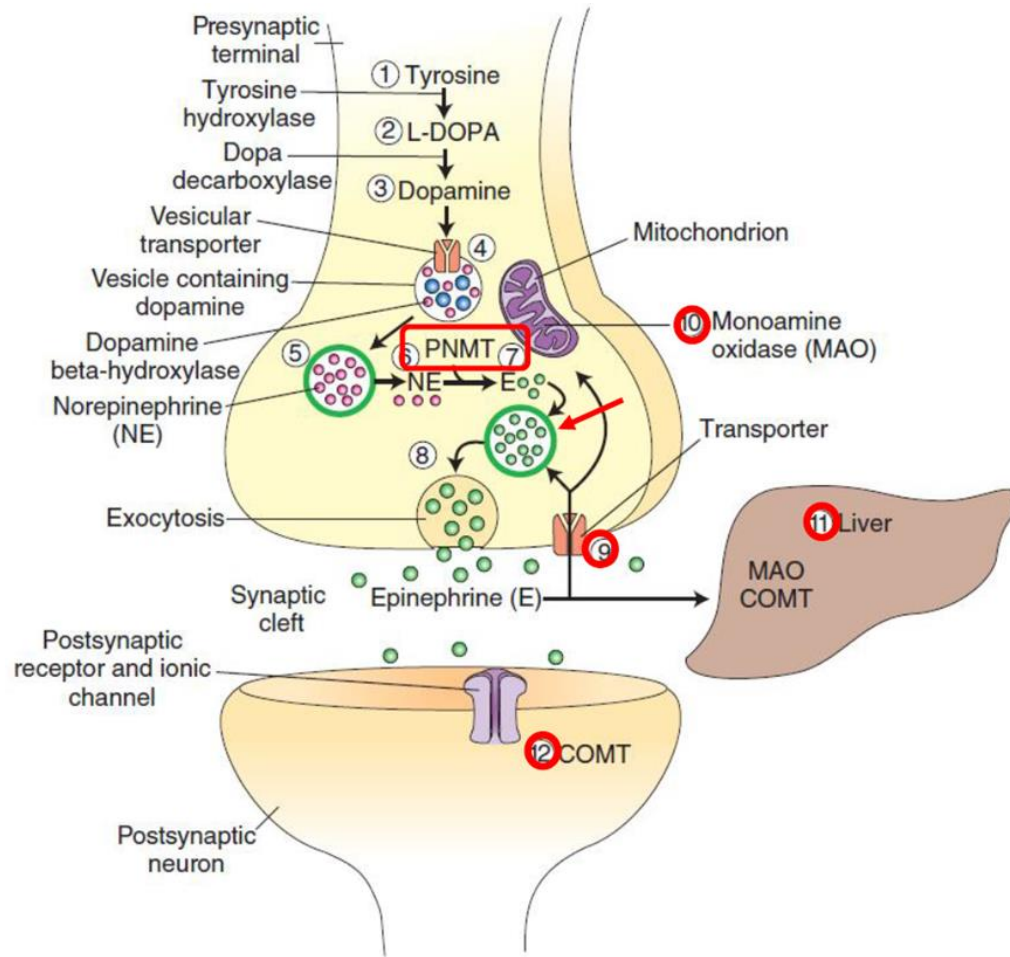
3. Reuptake-2: actively transports NE into postsynaptic cell for metabolism by COMT

4. Remaining synaptic NE diffuses and is absorbed by blood for peripheral metabolism

Steps involved in the synthesis and release of norepinephrine.

COMT, catechol-O-methyltransferase; L-DOPA, levodopa.

Synthesis and Removal of catecholamines: Epinephrine

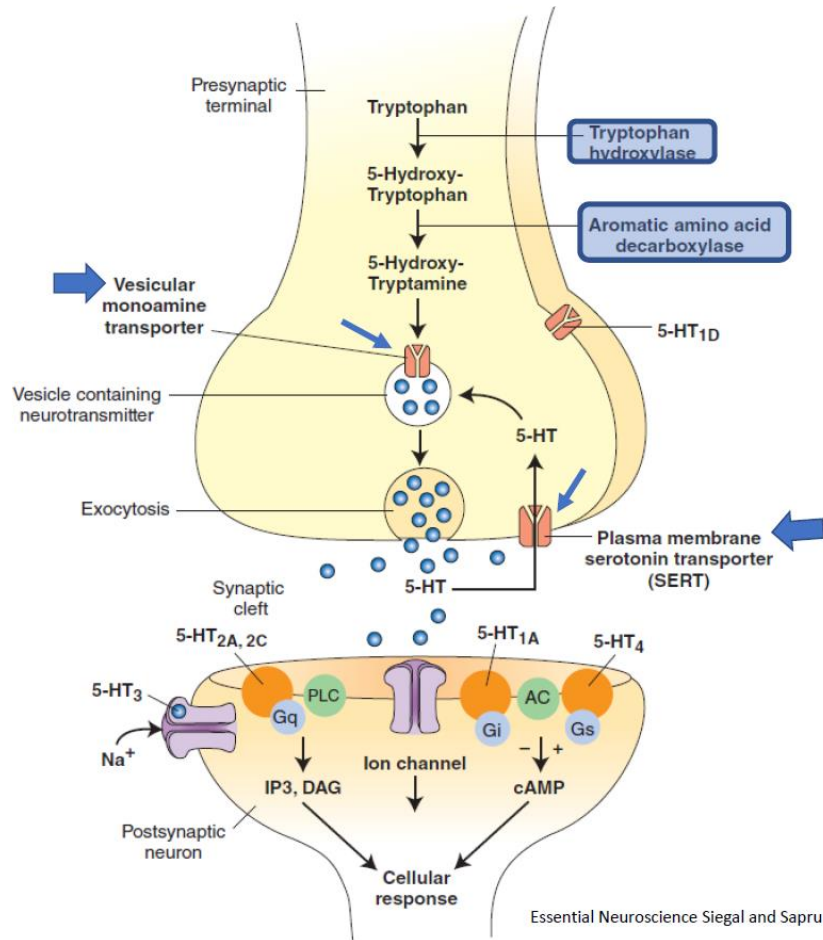


Catecholamines: Epinephrine

1. NE in vesicles leaks into the cytoplasm where it is converted to epinephrine by **phenylethanolamine-N-methyltransferase (PNMT)**
2. Reuptake-1 actively transports epinephrine into presynaptic neuron
Some E reloaded into vesicles
Remainder metabolized by MAO
3. Reuptake-2 actively transports epinephrine into postsynaptic cell for metabolism by COMT
4. Remaining synaptic epinephrine diffuses and is absorbed by blood for peripheral metabolism

Steps involved in the synthesis and release of epinephrine. *COMT*, catechol-O-methyltransferase; *L-DOPA*, levodopa; *PNMT*, phenylethanolamine-N-methyltransferase.

Synthesis and Removal of Indolamines: Serotonin



- Tryptophan hydroxylase converts cytoplasmic tryptophan to 5-hydroxytryptophan
- Aromatic amino acid decarboxylase converts 5-hydroxy-tryptophan to serotonin (5-hydroxy-tryptamine or 5-HT)
- Serotonin actively transported into vesicles for storage and then release
- Synaptic serotonin can undergo reuptake or metabolism by monoamine oxidase to 5-hydroxy-indoleacetyldehyde
- Aldehyde dehydrogenase then converts 5-hydroxy-indoleacetyldehyde to 5-hydroxy-indoleacetic acid for urinary excretion

Steps involved in the synthesis and release of serotonin (5-HT). The distribution of some 5-HT receptors on different components of the serotonergic synapse is also shown.

AC, adenylate cyclase; cAMP, cyclic adenosine monophosphate; DAG, diacylglycerol; Gi, Gs, Gq, different G proteins; IP₃, inositol triphosphate; PLC, phospholipase C; 5-HT_{1A}, 5-HT_{1D}, 5-HT_{2A}, 5-HT_{2C}, 5-HT₃, 5-HT₄, different 5-HT receptors.

Essential Neuroscience, 4e
Allan Siegel, Hreday N. Sapru,
Chapter 7: Neurotransmitters